

FlytBase AI Engineer Technical Assignment

Autonomous Drone Security Analyst Agent — Architecture & Verification Report

Candidate: Ketan Tiwari	Role: AI Engineer	Submission Date: October 2026
-------------------------	-------------------	-------------------------------

1. Executive Summary & Problem Approach

Modern enterprise security operations increasingly leverage autonomous **Drone-in-a-Box (DiaB)** systems to secure wide perimeters, critical logistics hubs, and commercial facilities. However, reviewing high-volume aerial footage manually induces surveillance fatigue and delayed emergency intervention. This project delivers an autonomous **Drone Security Analyst Agent** that ingests synchronized flight telemetry and video streams, performs real-time Vision-Language Model (VLM) reasoning, tracks object recurrence over time, enforces configurable facility security policies, and provides natural-language cross-domain frame search.

2. Architecture & Tool Justifications

2.1 VLM Selection: Why CLIP vs. BLIP?

A critical architectural requirement is selecting the appropriate vision-language paradigm for aerial surveillance:

Dimension	Salesforce BLIP	OpenAI CLIP	Our Hybrid Implementation
Primary Role	Generative dense image captioning.	Contrastive text-image alignment.	BLIP generative captions + CLIP/Vector search.
Key Strength	Generates detailed descriptive sentences for visual analysis.	Excels for robust classification and discrimination.	Combines both: BLIP for human-readable security logs while enabling CLIP for precise classification.
Output Type	Free-form natural language strings.	Dense 512-dim embedding vectors.	Structured Pydantic objects with captions and vector embeddings.

2.2 Dual-Storage Architecture (SQLite + ChromaDB)

To deliver robust cross-domain indexing, we implemented a dual-storage paradigm:

- **Relational (SQLite):** Stores structured telemetry (altitude, GPS, speed, battery), zone keys, and an immutable audit trail of security alerts.
- **Vector Store (ChromaDB):** Indexes frame visual captions and entity metadata into a dense vector index, supporting semantic queries like *'show all truck events'*.

3. Context Management & Security Rules Engine

A standout requirement of the assignment is contextual memory across frames: *'a blue Ford F150 entered twice today'*. The stateful ContextManager tracks entity signatures across missions, accumulating dwell times and distinct visit counts:

- **Loitering Detection Rule:** Tracks continuous presence in access zones. If dwell time exceeds 20s, a critical loitering alert is dispatched.
- **Curfew Enforcement Rule:** Automatically activates between 22:00 and 06:00, escalating unauthorized human or vehicle presence.
- **Recurrence Tracking Rule:** Detects vehicles returning to property grounds after prior departure, triggering immediate multi-visit security notices.

4. Target Output Verification & Test Results

Assignment Requirement	Target Output Pattern	System Verification Status
Operational Log	'Blue Ford F150 spotted at garage, 12:00.'	VERIFIED (Logged during 12:00 Midday Patrol)
Real-Time Alert	'Person loitering at main gate, 00:01.'	VERIFIED (Triggered at 00:01 Midnight Patrol)
Recurrence Tracking	'a blue Ford F150 entered twice today'	VERIFIED (Rule triggered upon visit #2 at 00:02)
Cross-Domain Search	'show all truck events'	VERIFIED (ChromaDB retrieves Frames 1, 2, and 7)
Bonus 1: Video Summary	1-Sentence Executive Debrief	VERIFIED (Automated post-patrol synopsis generated)
Bonus 2: LangChain Q&A	Follow-up agent dialogue	VERIFIED (Conversational tool-calling agent active)

5. Future Enhancements (Unconstrained Submission)

If submission timelines were unconstrained, production deployment would be expanded with:

- Edge Model Quantization (INT4 / TensorRT):** Deploying compact VLMs directly on drone companion boards (e.g. NVIDIA Jetson Orin Nano).
- Thermal Infrared Fusion:** Dual-sensor RGB/thermal overlay to detect human body heat through foliage or nighttime camouflage.
- Autonomous Docked Fleet Swarms:** Coordinating multiple DiaB stations via LangGraph to maintain persistent 24/7 airspace coverage during battery recharge cycles.

6. AI-Assisted Engineering Workflow

In accordance with assignment guidelines, AI-assisted tools (Cursor, Claude Code, Antigravity) were actively leveraged throughout development:

- **Scaffolding & Architecture:** AI assisted in designing decoupled Pydantic schemas and pipeline interfaces.
- **LangChain Integration:** AI generated baseline tool definitions, which were subsequently refined with security domain logic.
- **Automated QA Generation:** AI assisted in generating dynamic edge-case scenarios, achieving 100% pass rates across 9 automated pytest test suites.